(Please write your Enrolment No. immediately) Enrolment No. _____

MID TERM EXAMINATION

B. tech Second Semester, April, 2024

Paper Code: BS-104 Subject: Applied Chemistry

Time: 1 ½ Hrs. Max. Marks: 30

Note: Attempt Q. No. 1 which is compulsory and any two more questions from remaining.

Q. No.	Question	Max. Marks	CO(s)
1 (a)	A sample of coal c[illegible]tains C=63%, O=30%, H=6%, S=0.5%, N=0.2% and Ash=0.3%. Calculate the GCV of coal by Dulong's formula.	2	C[illegible] 1
(b)	What is Sweetening of Petrol (with reaction)?	2	CO-1
(c)	With the help of Phase Rule, find the Degree of Freedom of the following systems: (i) Ice & water in equilibrium (ii) Ag & Pb at the Eutectic point (iii) Saturated solution of NaCl (iv) Solid Ag & Soln. of Ag & Pb.	2	CO-2
(d)	Classify the following into Plastics, Elastomers & Fibres: (i)Nylon-6 (ii) HDPE (iii) Natural Rubber (iv) PVC	2	C0-2
(e)	Giv Composition, properties (any 2), Uses (any 2) of Water Gas & Producer Gas.	2	CO-1
Q 2 (a)	Describe how the GCV & NCV of a Solid fuel is determined using a Bomb Calorimeter (with diagram). Also give the formula of GCV with corrections involved.	5	CO-1
(b)	A gaseous fuel has the following composition by volume – CH_4=6%, CO=24%, CO_2=10%, H_2=10%, N_2=50%. If 30% excess air is supplied, calculate the volume of air supplied for 1m^3 of the fuel.	3	CO-1
(c)	Differentiate between Thermal Cracking & Catalytic Cracking.	2	CO-1
Q 3 (a)	With the help of a well labelled diagram explain the Sulphur System. Also explain why in a Sulphur System all the four phases cannot exist in equilibrium.	5	CO-2
(b)	Explain about Eutectic System, Eutectic mixture & Eutectic point.	3	CO-2
(c)	Write about the following polymers (i) Conducting polymers (II) Biodegradable Polymers (iii) LDPE (on any 2).	2	C0-2
Q 4.	Attempt any 2 parts		
(a)	What is the reason for Knocking in IC Engine. Define Octane No. & Cetane No.	5	CO-1
(b)	2.12g of the coal sample was Kjehldahlised and NH_3 gas thus evolved was absorbed in 30ml of 0.1N H_2SO_4. After absorption the excess acid required 10ml of 0.1N NaOH for complete neutralisation. Determine the percentage of Nitrogen.	5	CO-1
(c)	Explain about the Binary System with Congruent melting point & how it is different from Eutectic System.	5	CO-2
(d)	Classify the Polymers on the basis of Tacticity, polymerisation mechanism & on the basis of origin.	5	CO-2